PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Digital Platform for Centralized Alumni Data Management and Engagement

PROJECT REPORT

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IN

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "**Digital Platform for Centralized Alumni Data Management and Engagement**" is a Bonafide work of "D. Siddhardha Varma (20221CSE0412), Srinath K S (20221CSE0556), M. Venkata Riteesh (20221CSE0402)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING during 2025-26.

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DECLARATION

We the students of final year B. Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named D. Siddhardha Varma, Srinath K S, M. Venkata Riteesh, hereby declare that the project work titled **"Digital Platform for Centralized Alumni Data Management and Engagement"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B. Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

Alumni networks represent one of the most strategically valuable and yet most persistently mismanaged assets of educational institutions. Graduates who have built careers across diverse industries possess invaluable knowledge, industry connections, mentorship potential, and institutional goodwill that can significantly benefit current students and institutional development — but only if appropriate mechanisms exist to maintain meaningful engagement. Conventional approaches to alumni management, based on fragmented databases, spreadsheets, manual communication channels, and informal social media groups, are wholly inadequate for sustaining the level of engagement needed to realize this potential. The result is a persistent gap between the theoretical value of alumni networks and their practical impact on institutional outcomes.

This project addresses this challenge by designing, implementing, and validating a comprehensive Digital Platform for Centralized Alumni Data Management and Engagement. The platform integrates six core functional modules — Alumni Data Management, Alumni-Student Interaction and Mentorship, Job Referral Services, Event Management, Administrative Monitoring, and an Analytics Engine — into a single, secure, scalable web-based ecosystem that serves three distinct user roles: Alumni, Students, and Administrators.

The system is built on a four-layer architecture comprising a Presentation Layer (React.js, Bootstrap 5), an Application Layer (Node.js, Express.js, RESTful APIs), a Data Layer (MySQL centralized relational database), and an Analytics Engine for engagement metrics computation. Security is implemented through a comprehensive framework encompassing Role-Based Access Control (RBAC), JSON Web Token (JWT) authentication, HTTPS/SSL encryption, bcrypt password hashing, input validation, and parameterized database queries to prevent common web application vulnerabilities.

Development followed a hybrid V-Model/Agile methodology that combined the structured verification and validation discipline of the V-Model with the iterative, feedback-driven flexibility of Agile sprint development. This combination ensured both technical rigor and user responsiveness throughout the development process. The system was developed across a twelve-week timeline from January to April 2026, organized around ten development sprints.

The platform aligns with multiple United Nations Sustainable Development Goals including SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 10 (Reduced Inequalities). The results of this project demonstrate that a comprehensive, integrated digital alumni platform can be successfully developed within the resource constraints of a final-year academic project and that such a platform has the potential to significantly enhance alumni-institution relationships, improve student career outcomes, and support data-driven institutional development.